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A 3-item short version of the fear of missing out (FoMO) scale for trait and time-intensive designs: Measurement invariance and psychometric properties

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ABSTRACT

Research on Fear of Missing Out (FoMO) has gained prominence alongside the rise of digital connectivity and social media, which offer individuals continuous access to others' experiences in real time. Importantly, it has been increasingly recognised as an important psychological construct, closely associated with various problematic digital behaviours. As the fields of media psychology and human-computer interaction increasingly adopt time-intensive methodologies (e.g., daily and weekly studies) to capture dynamic experiences of technology use, there is a growing need for brief yet psychometrically-sound measures of FoMO suitable for such designs. The present work reports the development of a 3-item FoMO scale (FoMO-3) and evaluates its psychometric properties across trait and time-intensive methodologies such as daily diary studies and weekly diary studies. Across five independent samples (total N=1,258), we used structural equation modelling to evaluate the model fit and psychometric properties of trait-like and state-like versions of the FoMO-3. We found that across all assessments, the FoMO-3 consistently demonstrated high internal consistency. In Study 1, the FoMO-3 demonstrated strict measurement invariance and homogeneity of latent means and variances across sex. In Studies 2 and 3, strong time invariance was established across 13 consecutive weeks and 7 consecutive days respectively. Consistent with prior research, FoMO-3 scores were also positively associated with problematic smartphone use. These findings demonstrate that the concise, three-item FoMO-3 can capture the FoMO construct with strong reliability and validity, without compromising psychometric rigour.

1. Introduction

Research on Fear of Missing Out (FoMO)—a pervasive apprehension that others might be having rewarding experiences when one is absent [85]—has shown that FoMO is an important psychological construct associated with numerous important life outcomes [38,105]. Originally emerging within the context of increased social media use and online connectivity, FoMO has been shown to be closely associated with problematic digital behaviours, such as compulsive social media use, phubbing, problematic smartphone use, and risky loot box consumption [8,13,27,34,51,59,78,81,116]. Beyond the digital domain, FoMO has also been linked to significant psychological distress, such as increased anxiety, depressive symptoms, and loneliness [7–9,69,81]. Additionally, FoMO has been associated with disruptions in daily functioning, including poorer sleep quality [1], reduced concentration and productivity [5,6,16,97], and increased engagement in risky behaviours such as alcohol consumption [92,93] and unsafe driving [14,73]. Given the extensive and multifaceted impact of FoMO, accurately measuring and addressing FoMO with valid and reliable tools is crucial.

1.1. *Existing measures of FoMO*

Primarily, FoMO is most commonly assessed using the original 10-item scale developed by Przybylski et al. [85], with systematic reviews of the literature having found that this specific measure has been used in 62–94 % of studies of FoMO from 2013 to 2024 inclusive [17,40,

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115]. The original 10-item scale, which we henceforth refer to as the FoMO-10, captures the trait-like extent to which individuals experience fear of missing out on events, experiences, or social activities (e.g., "I fear others have more rewarding experiences than me", "I get worried when I find out my friends are having fun without me", "Sometimes, I wonder if I spend too much time keeping up with what is going on"). While the FoMO-10 serves as a foundational tool for FoMO research, its trait-like nature limits researchers' ability to capture within-person fluctuations. Indeed, researchers have recognised the need for a more flexible and adaptable measure across different contexts.

Interest in examining FoMO in time-intensive study designs has grown in recent years. For instance, researchers have attempted to measure FoMO four or even five times a day in experience sampling method and ecological momentary assessment designs (e.g. [44,76]). Owing to the need for efficient assessments, a shortened measure of FoMO has been used in such cases, consisting of a single item: "Do you experience FoMO?" [91]. The development of this single-item version is notable as it offers multiple benefits over the 10-item scale, requiring less cognitive workload on participants [4] and less administration time for both participants and researchers [45,63,84]. Thus, the single-item version is highly practical for study designs that involve multiple measures administered repeatedly. However, single-item measures are limited in their capacity to assess psychometric properties, such as internal consistency, and are more susceptible to measurement error [30,33,35,42,82].

Importantly, as research in media psychology and human-computer interaction increasingly adopts time-intensive methodologies (e.g., daily or weekly diary studies) to examine technology use at the state level (e.g. [51,54,77,83,95]), the need for a psychometrically sound yet concise measure that can capture these within-person fluctuations in FoMO has become especially salient. To this end, Holte [55] developed an 8-item state FoMO scale designed to capture state fluctuations in the severity of FoMO, drawing on Compensatory Internet Use Theory (Kadefelt-Winther, 2014). While Holte's [55] scale contributes to the literature by operationalising FoMO as a state construct, its emphasis is primarily on special or unique events (e.g., "I fear I am absent from a special moment"), which makes it particularly suited for capturing event-based FoMO. However, the FoMO-3 developed in the present work was designed to target the core interpersonal anxieties that are flexible for typical day-to-day contexts.

Notably, there has been an attempt to develop a measure of FoMO that taps onto both its trait and state components concurrently, namely the Trait-State FoMO Scale [110]. Although the Trait-State FoMO Scale acknowledges both dispositional and situational components of FoMO, its state subscale is overly focused on online activity and the use of social networking sites (e.g., "It is important that I have a say about the latest issues in my online social networks"). While these items capture digital expressions of FoMO, they might miss more interpersonal and everyday concerns across both online and offline domains that the FoMO-3 was developed to capture in the present work.

1.2. Measurement invariance

Regardless of the choice of measure, an important issue is whether such measures function equivalently across groups. Prior studies have reported mixed findings on sex differences in levels of FoMO; some have found that males have higher levels of FoMO than females (e.g. [87]), while others have found the reverse (e.g. [103]), and some have even found no evidence for a meaningful difference between males and females (e.g. [98]). However, such comparisons are built on the assumption that these measures operate equivalently across both males and females. To examine whether observed differences reflect true variation in FoMO across groups, rather than due to differences in measurement, researchers must first test for measurement invariance across groups [23].

Currently, only a small number of studies have investigated the issue of sex invariance, and the limited evidence that has been accrued for the FoMO-10 has been mixed. For example, the Italian version [20] of the FoMO-10 has demonstrated strong invariance, while the Persian version [21] and the original English version [36] have demonstrated only configural invariance. Focusing on the original English version of FoMO-10 specifically, evidence on measurement invariance is still relatively scarce, despite its widespread use. These gaps and inconsistencies highlight the need to examine measurement invariance more rigorously.

Beyond the existing issues, confirmation of measurement invariance in a shortened version is important to establish, as psychometric properties can change when scales are modified in any way. In addition, since the FoMO-3 is intended for time-intensive studies, evaluating its time invariance is equally important to assess whether changes in scores over time reflect true within-person fluctuations in FoMO, rather than a shift in item properties over time. Yet, despite the use of FoMO measures in both panel studies (e.g. [49,68]) and ecological momentary assessment studies (e.g. [44,76]), formal tests of measurement invariance over time remain limited.

1.3. The current work

The current landscape of FoMO measurement reflects differences in conceptualisation, with some scales assessing the construct as a dispositional trait, others as a transient state, and some as both. Yet, existing measures lack rigorous validation and evidence of equivalence across groups. With the growing use of time-intensive designs, the need for a brief, robust measure of FoMO has become especially pressing.

Thus, the present work seeks to provide a balanced measure of FoMO that captures both trait-like stability and state-like fluctuations, while remaining brief and psychometrically sound for use in time-intensive designs. To this end, we first conducted a pilot study using the original FoMO-10 [85] to identify items for a shortened version of the measure. This process yielded FoMO-3, which was psychometrically evaluated in the subsequent studies. We investigated FoMO along a trait-like to state-like continuum. Study 1 focused on establishing properties of trait-like FoMO using a more dispositionally-worded version of the FoMO-3. Trait-like FoMO represents a stable and enduring characteristic of an individual that remains relatively unchanged over time and can be compared between individuals. Studies 2 and 3 focused on a more state-like component of FoMO, which refers to a more dynamic experience of FoMO that varies depending on time and situational factors within individuals [5,74,85]. Specifically, Study 2 examined the properties of a weekly FoMO-3 measure, which reflects patterns of FoMO on a week-to-week basis using weekly diary data, while Study 3 investigated properties of a daily FoMO-3 measure, which captures fluctuations in FoMO experienced each day using daily diary data. Table 1 provides an overview of all the studies, including their samples and main aims.

2. Transparency and openness

The relevant materials, data, and code required to reproduce our analyses have been made publicly available on ResearchBox (#3841; https://researchbox.org/3841). All analyses and procedures were conducted in R version 4.5.0 [89]. Data cleaning and manipulation were performed using *dplyr* version 1.1.4 [112] and *tidyr* version 1.3.1 [113]. Descriptives and tau-equivalent reliabilities and corresponding 95 % CIs for composite measures were extracted using *psych* version 2.5.6 [90]. Main analyses were conducted using *lavaan* version 0.6.19 [96] and *semTools* version 0.5.7 [60]. Visualisations were created using *ggplot2* version 3.5.2 [111] and *ggpubr* version 0.6.1 [61].

Table 1
Overview of studies and samples.

Study	Sample	Sample Size	Data Structure	Aim of Study
Pilot Study	25 % subsample from two daily diary projects (see [47, 108])	124	Cross-sectional	Develop a shortened 3-item scale from FoMO-10
Study 1	75 % subsample from two daily diary projects (see [47, 108])	372	Cross-sectional	Evaluate the psychometric properties of FoMO-3 as a trait-like measure
Study 2	Sample from a weekly diary project (see [51])	250	13-week diary	Evaluate the psychometric properties of FoMO-3 as a weekly state-like measure
Study 3	Sample from two daily diary projects (see [46,53,70, 80]).	512	7-day diary	Evaluate the psychometric properties of FoMO-3 as a daily state-like measure

3. Pilot study: item selection

3.1. Method

3.1.1. Sample and design

In this study, data was drawn from a total of 124 unique individuals, a subset (25 %) of 496 individuals who participated in the baseline sessions of two larger daily diary projects [47,108]. Participants were recruited via convenience sampling from local universities in Singapore. Data collection was approved by a local Institutional Review Board, and all participants provided informed consent to participate prior to the start of data collection. Participants received either course credit or cash compensation (up to SGD$72 or SGD$120 respectively in each project) for their participation in the broader diary studies. A summary of relevant descriptives is provided in Table 2. There was no missing data.

Table 2
Participant-level descriptive statistics.

Characteristic	*M* (*SD*) or %		Observed Range	Theoretical Range
Demographics				
Sex (% female)	67.74 %			
Race (% Chinese)	79.03 %			
Age	21.52	(1.81)	18–25	
Socioeconomics				
Monthly Household Income	3.59	(1.52)	1–6	1–6
Ladder Rank	6.29	(1.31)	2–9	1–10
FoMO-10				
Item 1: “I fear […] others […] rewarding experiences than me.”	2.79	1.20	1–5	1–5
Item 2: “I fear […] friends […] rewarding experiences than me.”	2.73	1.25	1–5	1–5
Item 3: “I get worried […] friends are having fun without me.”	2.82	1.30	1–5	1–5
Item 4: “[…] anxious […] what my friends are up to.”	2.16	1.24	1–5	1–5
Item 5: “It is important […] understand […] “in jokes”.”	2.83	1.23	1–5	1–5
Item 6: “[…] keeping up with what is going on.”	2.50	1.28	1–5	1–5
Item 7: “[…] bothers […] miss an opportunity […] friends.”	2.96	1.25	1–5	1–5
Item 8: “[…] important […] share the details online […].”	2.30	1.20	1–5	1–5
Item 9: “[…] miss out […] get-together […] bothers me.”	3.09	1.33	1–5	1–5
Item 10: “[…] keep tabs on […] friends are going.”	2.20	1.17	1–5	1–5

Note. $N = 124$.

3.1.2. Measures

3.1.2.1. Trait-like FoMO. Trait-like FoMO was measured by the full, original 10-item Fear of Missing Out Scale [85]. The original phrasing of all items was retained in the current work. Participants rated their trait-like level of FoMO on a 5-point scale (1=*Not at all true of me*, 5=*Extremely true of me*).

3.1.2.2. Demographics and socioeconomics. Sex was measured as male or female (0=*Male*, 1=*Female*), while race was measured based on Singapore’s predominant ethnic classification method (1=*Chinese*, 2=*Malay*, 3=*Indian*, 4=*Others*; [29]). Age was measured in years. Socioeconomic status was measured both in terms of an objective measure and a subjective measure. Objective socioeconomic status was assessed as monthly household income on a 6-point scale (1=*less than $2000*, 2=*$2000–$5999*, 3=*$6000 –$9999*, 4=*$10,000–$14,999*, 5=*$15,000–$19,999*, 6=*more than $20,000*). Subjective socioeconomic status was assessed as a ladder rank using the MacArthur scale (1=*lowest subjective socioeconomic status*, 10=*highest subjective socioeconomic status*; [2]), where participants were asked to indicate their perceived position within their community by selecting a rung on a symbolic ladder.

3.1.3. Analytic plan

3.1.3.1. Item selection. We entered all 10 items of the FoMO scale as manifest variables of an underlying unidimensional latent factor of FoMO in a confirmatory factor analysis. Items were evaluated against two criteria. Firstly, the factor loadings of the items should be relatively high (i.e., $\lambda \geq .60$; [22,58]; Hair et al., 2010). Secondly, the items should retain conceptual coherence to represent FoMO adequately [28]. To this end, we examined the factor loadings of all 10 items and retained the new 3-item scale (i.e., FoMO-3) for further evaluation in the rest of the Pilot Study, and the present work. In addition, we evaluated the comparability of the FoMO-3 with the original FoMO-10 by computing the simple correlation between the mean score derived from the FoMO-10 and the mean score derived from the FoMO-3.

3.1.3.2. Psychometric properties. We re-fit a confirmatory factor model using the FoMO-3 to preliminarily evaluate its psychometric properties.

3.1.3.2.1. Internal consistency. We evaluated internal consistency of FoMO-3, as assessed by two key coefficients: tau-equivalent (α; [32]) and congeneric (ω; [39,48]) estimates. These coefficients measure whether the extent to which the included items are, as a whole, unidimensionally related to the same underlying construct. Since α assumes tau-equivalence (i.e., equal factor loadings of all items in a factorial model; [25]) and such an assumption is not always tenable, it is essential to calculate congeneric estimates as well. Tau-equivalent internal consistencies were included in order to allow researchers to compare estimates across studies, given that α is still frequently used [25]. More importantly, α values are useful for researchers who wish to interpret mean scores of the FoMO-3 instead of latent factor scores, as mean scores assume unidimensionality and equal factor loadings across items. Hence, reporting α is relevant in this context. Internal consistency was considered good if the estimate was above 0.70 [66,106].

3.1.3.2.2. Average variance extracted. We calculated the average variance extracted (AVE) of the FoMO factor to determine the average proportion of variance in the observed items explained by trait-like FoMO. An AVE of 0.50 or higher indicates that the latent factor explains an acceptable level of variance in the manifest items [22,41].

3.2. Results

3.2.1. Item selection

We found that seven out of the 10 items of the full, original FoMO scale had factor loadings which were ≥.60 (Fig. 1). Of the seven items, we selected the following three items: “I get worried when I find out my friends are having fun without me” (λ=0.755, p<.001), “I get anxious when I don’t know what my friends are up to” (λ=0.760, p<.001), and “It bothers me when I miss an opportunity to meet up with friends” (λ=0.723, p<.001). The items were selected as they exceeded the threshold of λ≥.60, and captured a general, and recurring form of missing out that can arise across many daily and weekly contexts. The three items adequately reflect fears, worries, and anxieties people may have in relation to being in or out of touch with events and experiences happening across their social circles ([51]; Przybylski, 2013).

Further, the simple correlation between the mean score of the FoMO-10 and the mean score of the FoMO-3 was positive and nearly perfect at r=0.912, 95 % CI=[.877, 0.937] (Fig. 2). Taken together, the results show that these three items balance psychometric strength with theoretical coverage, while remaining practical for daily and weekly diary assessments.

3.2.2. Psychometric properties

Using the three retained items, we found that both tau-equivalent (α=0.786) and congeneric (ω=0.793) estimates of internal consistency exceeded 0.70, which denotes good inter-item consistency. Average variance extracted (AVE=0.566) was above the acceptable threshold of 0.50, which shows that the new latent factor of FoMO accounted for more than half of the variance in scores on the three retained items. Overall, we found that the newly-developed FoMO-3 exhibits good psychometric properties in the initial sample.

4. Study 1: trait-like FoMO

4.1. Method

4.1.1. Sample and design

In this study, data was drawn from baseline data of a total of 372 unique individuals (109 males and 263 females), the remaining subset (75 %) of 496 individuals who participated in the baseline sessions of two larger daily diary projects [47,108] who were not used in the previous Pilot Study. Participants received either course credit or cash compensation (up to SGD$72 or SGD$120 respectively in each project) for their participation in the broader diary studies. There was no missing data. A summary of baseline descriptives is provided in Table 3.

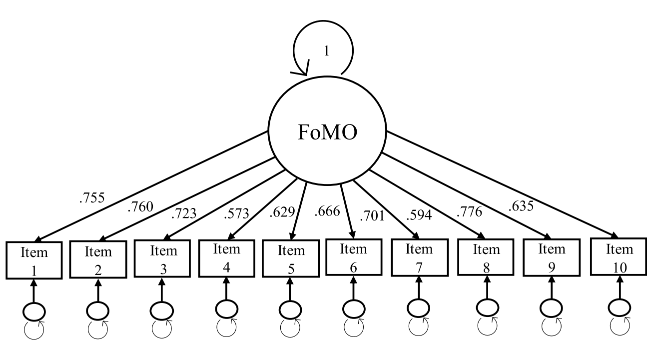

Fig. 1. Factor Structure of the 10-item FoMO Scale.
Note. All values shown are standardised, including latent variance which was fixed to 1 for model identification. Items 1, 2, and 3 were selected to form the 3-item short version (FoMO-3) that was used in subsequent analyses. All factor loadings, intercepts, and error variances were statistically significant (all ps<0.001). Intercepts are not shown.

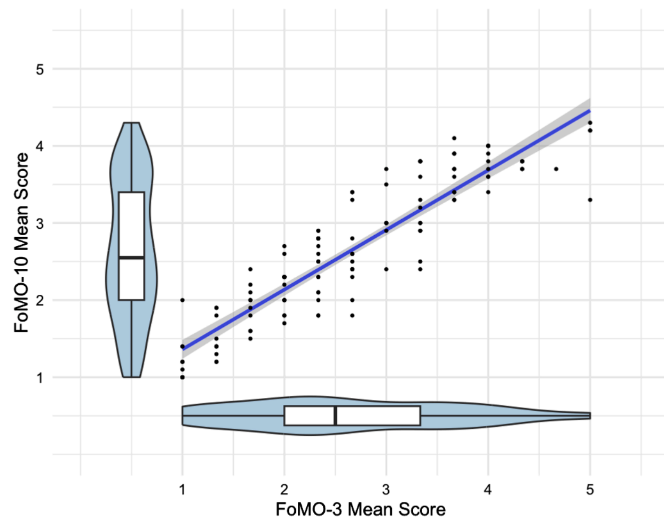

Fig. 2. Simple correlation between FoMO-10 mean score and FoMO-3 mean score.
Note. N = 124. Scatterplot with violin and boxplots illustrating the simple correlation between the mean scores of the FoMO-10 and FoMO-3 scales.

4.1.2. Baseline measures

4.1.2.1. Trait-like FoMO. Trait-like FoMO was measured using the three items selected in the Pilot Study (i.e., “I felt worried when I found out my friends were having fun without me, “I felt anxious without knowing what my friends were up to”, “I felt bothered due to a missed opportunity to meet up with friends”). The original phrasing of the three items was retained in the current work. Participants rated their trait-like level of FoMO on a 5-point scale (1=*Not at all true of me*, 5=*Extremely true of me*).

4.1.2.2. Demographics and socioeconomics. Demographic and socioeconomic variables were measured as in the Pilot Study.

4.1.2.3. Big five personality traits. Participants’ big five personality traits were measured using the 30-item Big Five Inventory-2-Short [102]. Out of the 30 items, six items were used to measure each facet of personality: openness to experience (e.g., “Is fascinated by art, music, or literature”; α=0.68 [.63, 0.73]), conscientiousness (e.g., “Tends to be disorganised”; α=0.74 [.69, 0.77]), extraversion (e.g., “Is full of energy”; α=0.79 [.75, 0.82]), agreeableness (e.g., “Assumes the best about people”; α=0.71 [.66, 0.75]), and neuroticism (e.g., “Is emotionally stable, not easily upset”; α=0.84 [.81, 0.86]). Participants were asked how much they agreed these characteristics applied to them on a 5-point scale (1=*Strongly disagree*, 5=*Agree strongly*).

4.1.2.4. Trait positive and negative affectivity. Trait levels of positive and negative affectivity were measured using the PANAS-X scale [109]. Participants rated their trait level of positive and negative emotions on a 5-point scale (1=*Very slightly or not at all*, 5=*Extremely*) during the respective baseline questionnaires. Positive affect was assessed using 10 items (e.g., “active”, “determined”, “proud”; α=0.87 [.85, 0.89]), while negative affect was assessed using 10 items (e.g., “afraid”, “nervous”,

Table 3
Baseline descriptive statistics.

Characteristic	Male (N = 109)			Female (N = 263)			
	M (SD) or %		Observed Range	M (SD) or %		Observed Range	Theoretical Range
Demographics							
Race (% Chinese)	77.01 %			75.60 %			
Age	22.85	(2.00)	18–30	21.16	(1.66)	18–28	
Socioeconomics							
Monthly Household Income	3.09	(1.35)	1–6	3.24	(1.48)	1–6	1–6
Ladder Rank	5.99	(1.51)	2–8	6.07	(1.26)	3–9	1 –10
Big Five							
Openness	3.42	(0.68)	2.00–4.83	3.49	(0.72)	1.33–5.00	1–5
Conscientiousness	3.09	(0.73)	1.00–4.67	3.07	(0.70)	1.33–5.00	1–5
Extraversion	3.03	(0.85)	1.00–5.00	2.93	(0.79)	1.17–5.00	1–5
Agreeableness	3.49	(0.67)	1.00–4.83	3.55	(0.67)	1.67–5.00	1–5
Neuroticism	2.78	(0.90)	1.00–5.00	3.21	(0.83)	1.33–5.00	1–5
Trait Affect							
Positive Affect	3.18	(0.77)	1.20–4.90	2.97	(0.69)	1.00–4.40	1–5
Negative Affect	2.28	(0.90)	1.00–4.80	2.30	(0.78)	1.00–4.70	1–5
Smartphone Addiction	3.06	(0.98)	1.00–6.00	3.20	(0.92)	1.00–5.40	1–6
FoMO-3 (Trait Version)							
"[…] worried […] friends are having fun without me."	2.86	(1.33)	1–5	2.79	(1.30)	1–5	1–5
"[…] anxious […] what my friends are up to."	2.22	(1.25)	1–5	2.07	(1.17)	1–5	1–5
"[…] bothered […] missed opportunity to meet up with friends."	2.93	(1.25)	1–5	2.79	(1.20)	1–5	1–5

Note. N = 372.

"distressed"; α=0.90 [.88, 0.91]).

4.1.2.5. Smartphone addiction. Smartphone addiction was measured using the Smartphone Addiction Scale—Shortened Version [64]. Participants rated the extent to which they agreed with 10 statements reflecting excessive or compulsive smartphone use that interfered with daily activities (e.g., "Missing planned work due to smartphone use", "Having a hard time concentrating in class, while doing assignments, or while working due to smartphone use"; α=0.85 [.82, 0.87]) on a 6-point scale (1=*Strongly Disagree*, 6=*Strongly Agree*).

4.1.3. Analytic plan

As in the latter portion of the Pilot Study, in Study 1, we modelled the latent factor of FoMO as being unidimensionally manifested by the three items of the FoMO-3 in a confirmatory factor analysis.

4.1.3.1. Measurement invariance over sex. We tested for measurement invariance to ensure that the FoMO-3 operates in the same manner across sex. Measurement invariance across sex is held if the stable construct of FoMO is measured equivalently across both groups (i.e., males and females). In order for invariance to hold, at least two out of three of these criteria must be met: AIC [3,50,65] and/or BIC values are lower [72,99] and/or the change in CFI is <0.010 (i.e., ΔCFI<0.010; [18,23]).

Invariance was tested using four consecutive steps ([86,104]; Putnick & Bornstein, 2016): configural invariance, weak (or metric) invariance, strong (or scalar) invariance, and strict (or residual) invariance. At each step, the model from the previous step was compared to a new alternative model. The first step of configural invariance was tested by constraining the general factor structures to be the same across sex. Next, weak invariance was tested by comparing the configural invariance model to a model with equality constraints of factor loadings across groups (i.e., the relationship between latent FoMO and items is equal across sex). Strong invariance was then assessed by comparing the weak invariance model to a new model with additional equality constraints on intercepts (i.e., item scores are equal across sex when the level of the FoMO construct is set to zero). Lastly, strict invariance was tested by comparing the strong invariance model to an alternative model with the additional equality constraints of residual variance (i.e., any unexplained variance is equal across sex). If strict invariance was held, we can conclude that FoMO is measured identically across males and females [88].

4.1.3.2. Homogeneity of latent variances and means by sex. After establishing measurement invariance, tests for homogeneity of latent variances and means were conducted simultaneously on the final model, which passed measurement invariance testing. Specifically, we tested for equality of latent means to determine whether males and females exhibit similar latent levels of FoMO and equality of variances to assess whether the variance of the FoMO factor was consistent between males and females. We used the same criteria as measurement invariance testing (i.e., lower AIC, lower BIC, and/or ΔCFI<0.010). If the model met at least two of these criteria, we concluded that males and females do not differ in the latent means and latent variances in FoMO.

4.1.3.3. Psychometric properties. To ensure that the FoMO-3 is a valid measurement of trait-like FoMO, we assessed its psychometric properties in terms of internal consistency and average variance extracted, mirroring our analyses in the Pilot Study.

4.1.3.4. Correlations with FoMO. Next, we explored the relationship of latent FoMO with the following variables: age, objective (income) and subjective (ladder rank) socioeconomic status, big five personality traits (i.e., openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism), positive affect, negative affect, and smartphone addiction. Correlations were assessed using Pearson's r, following Funder and Ozer's [43] guidelines, where 05≤r<0.10 suggests a very small effect size, 0.10≤r<0.20 suggests a small effect size, 0.20≤r<0.30 suggests a medium effect size, 0.30≤r<0.40 suggests a large effect size, and an effect size of 0.40≤r < 1.00 suggests a very large effect size. Additionally, we investigated whether these correlations between FoMO and each of these variables differed by sex[1] by comparing two models: a constrained and an unconstrained model.

The unconstrained model allowed all variables to correlate freely across sex (i.e., correlations between variables across males and females were not forced to be equal), with the exception of the intercorrelations within the demographic variables (i.e., age, objective (income), and subjective (ladder rank) socioeconomic status), intercorrelations within

[1] Differences by race would also be potentially informative to explore. However, we could not examine race in the current work due to insufficient representation of minority race individuals (i.e., non-Chinese participants) in the dataset and a lack of theoretical justification for potential differences in correlations across race.

the affective variables (i.e., positive and negative affect), intercorrelations between the affect-adjacent personality variables and their respective valence of affect (i.e., extraversion with positive affect; neuroticism with negative affect [31,56,75]), and intercorrelations between psychopathology-adjacent variables (i.e., neuroticism and smartphone addiction [19,67]); which were constrained to equality across sex. This process ensured that any differences in correlations between FoMO and the variables were not confounded by variability in relationships within any variable. The constrained model forced all remaining correlations (that is, between FoMO and the other nine variables) to be equal across sex to test the hypothesis that the relationships between FoMO and the relevant variables are the same across males and females.

The two models were compared against each other using the same criteria as measurement invariance testing (i.e., lower AIC, BIC, or ΔCFI<0.010). This allowed us to identify the model with a better fit. If the constrained model showed a better fit, we can conclude that there were no significant sex differences in the relationship between FoMO and the other variables. Conversely, if the unconstrained model showed a better fit, we can conclude that there were significant sex differences in the relationship between FoMO and the other variables. Subsequently, using the model with the better fit, we interpreted the individual correlations between FoMO and each variable.

4.2. Results

4.2.1. Measurement invariance by sex

Weak invariance was tested by comparing the configural invariance model against a new model with factor loadings constrained to equality across sex. Examinations of CFI, AIC, and BIC values showed that weak invariance was held based on all three indices (Table 4). Next, strong invariance was tested by comparing the weak invariance model to a model with the additional equality constraint of item intercepts across sex. Examinations of CFI, AIC, and BIC values revealed that strong invariance was held based on the findings of CFI and BIC (Table 4). Afterward, we tested for strict invariance by comparing the strong invariance model to a model with the additional equality constraint of residual variances across sex. We found that strict invariance was supported based on the CFI, AIC, and BIC. In conclusion, the 3-item FoMO scale was found to be invariant across sex, that is, FoMO was measured identically across males and females using the three items.

4.2.2. Homogeneity of latent variances and means by sex

Using the strict invariance model, we then conducted homogeneity tests of latent means and variances by sex. This involved comparing the strict invariance model to an alternative model with an added constraint of equal variances and equal means across sex. At this step, we found no evidence of any significant differences in latent variances or latent means between sex based on CFI, AIC and BIC (Table 4). Hence, males and females exhibit a similar distribution of the FoMO construct. The final factor structure is shown in Fig. 3.

Table 4
Model fit indices and model comparison indices of trait-like FoMO model.

Model	RMSEA	SRMR	CFI	ΔCFI	AIC	BIC
Measurement Invariance						
Configural	–	–	–	–	3367	3437
Weak (vs. Configural)	<0.001	.008	>0.999	<0.001	3363	3426
Strong (vs. Weak)	<0.001	.012	>0.999	<0.001	3360	3415
Strict (vs. Strong)	<0.001	.015	>0.999	<0.001	3355	3398
Homogeneity						
Variances and Means (vs. Strong)	<0.001	.043	>0.999	<0.001	3353	3388

Note. $N = 372$ (female $N = 263$, male $N = 109$). RMSEA, SRMR, and CFI were not available for the configural model as the model was saturated.

4.2.3. Psychometric properties

For both males and females, both tau-equivalent (α_{male}=0.806; α_{female}=0.754) and congeneric (ω_{male}=0.703; ω_{female}=0.817) estimates of internal consistency exceeded 0.70, which denotes good inter-item consistency. Average variance extracted (AVE_{male}=0.514; AVE_{female}=0.556) was above the acceptable threshold of 0.50, which shows that trait-like FoMO accounted for more than half of the variance in item scores. Overall, the FoMO-3 exhibits good psychometric properties at the trait level.

4.2.4. Correlations with FoMO

First, we found that the constrained model (AIC=14,053, BIC=14,245, CFI=0.574) displayed a better fit compared to the unconstrained model (AIC=14,062, BIC=14,297, CFI=0.576). This suggests that there was no evidence of sex differences in the correlations between FoMO and the variables.

Upon further investigation of the individual correlations between each variable and FoMO (Fig. 4), trait-like FoMO showed significant zero-order correlations with agreeableness, extraversion, neuroticism, positive affect, negative affect, and smartphone addiction but not any of the other included variables (i.e., age, objective [income] and subjective [ladder rank] socioeconomic status, conscientiousness, openness to experience, nor positive affect). Specifically, FoMO was positively related to extraversion (r=0.119) and positive affect to a small extent (r=0.101), smartphone addiction (r=0.265) to a medium extent, negative affect (r=0.372) to a large extent, and neuroticism (r=0.454) to a very large extent such that people with higher levels of FoMO tended to be more extraverted, neurotic, have higher levels of trait positive and negative affect, and smartphone addiction than people with lower levels of FoMO. On the other hand, trait-like FoMO was negatively related to agreeableness (r=–.164) to a small extent, such that people with higher levels of FoMO tended to be less agreeable.

5. Study 2: weekly FoMO

5.1. Method

5.1.1. Sample and design

The sample for the study of weekly FoMO was drawn from a weekly diary study comprising an initial total of 257 individuals tracked over an academic semester (see [51]). Following the baseline questionnaire, participants were instructed to complete 14 online surveys administered weekly over 14 weeks. In the current study, only 13 weekly surveys, excluding the first survey administered prior to the start of the semester, were used to facilitate model simplification and mitigate potential confounding. Participants were allowed to respond to the survey from 12 a.m. on Saturday till 11.59 pm. on Sunday and were provided compensation of up to SGD$135 for their participation. Participants were included if they completed at least two weeks of weekly data collection (excluding the first week prior to the start of the semester), reducing the sample from 257 to 250 participants. This resulted in a total of 3018 observations of weekly data available for analysis. A summary of descriptive statistics is reflected in Table 5 and Fig. 5a.

5.1.2. Weekly measures

5.1.2.1. Weekly FoMO.

Weekly FoMO was measured using the same 3-item Fear of Missing Out Scale [85]. The scale was adapted to include a 1-week timeframe (i.e., "In the past week") at the beginning of the item to guide the specificity of responses, given that participants were made to respond once a week. Participants were asked to rate their general level of FoMO throughout the week on a 5-point scale (1=*Not at all true of me*, 5=*Extremely true of me*).

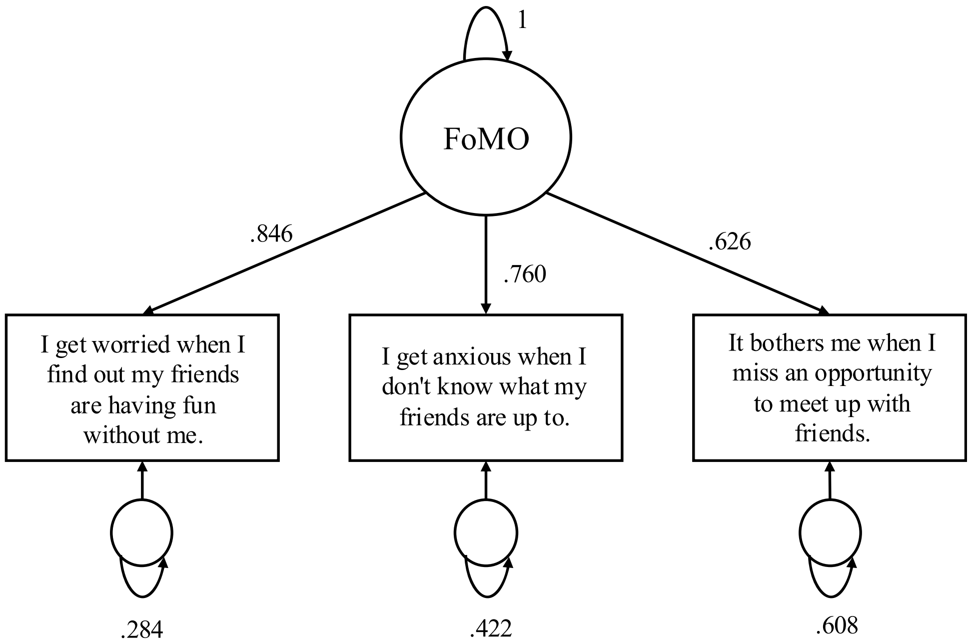

Fig. 3. Final factor structure of trait-like FoMO.
Note. Factor structure was identical for both males ($N = 109$) and females ($N = 263$). All values shown are standardised, including the latent variance which was fixed to 1 for model identification. All factor loadings, intercepts, and error variances were statistically significant (all $ps<0.001$). Intercepts are not shown.

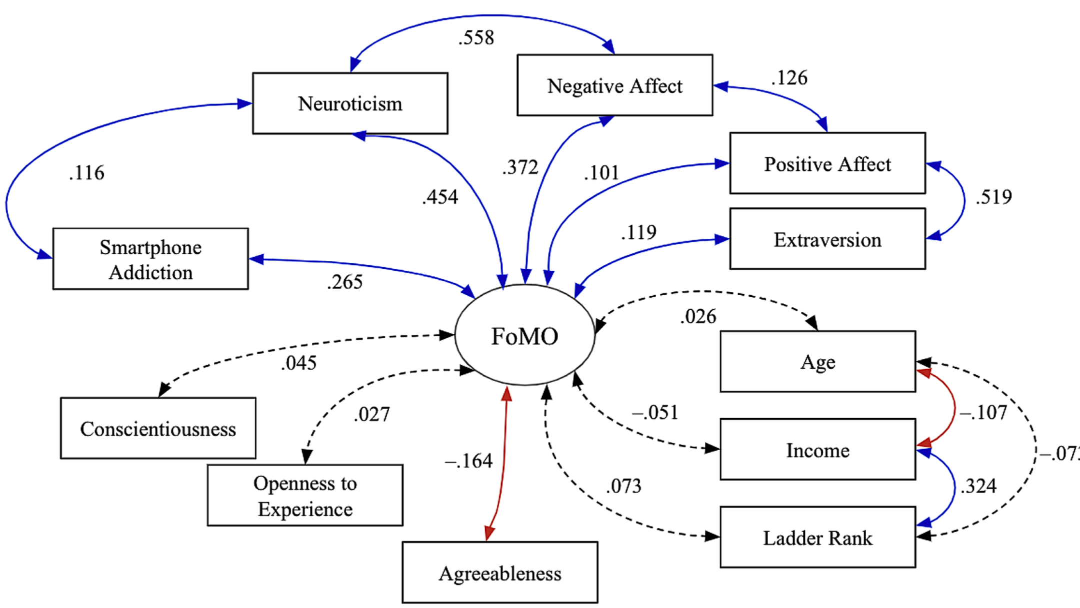

Fig. 4. Zero-order correlations of trait-like FoMO with other trait variables.
Note. $N = 372$ (female $N = 263$, male $N = 109$). All values refer to bivariate zero-order correlations (i.e., standardised covariances). Solid lines refer to statistically significant correlations, where blue lines refer to positive correlations and red lines refer to negative correlations. Dashed lines refer to non-significant correlations. The three manifest variables of FoMO are not shown.

5.1.2.2. Weekly positive and negative affectivity. Weekly positive and negative affectivity were measured using the 20-item PANAS-X scale [109]. Participants were asked how frequently they experienced the following positive and negative emotions on a 5-point scale (1=*Very slightly or not at all*, 5=*Extremely*) throughout the week. Positive affect was assessed using 10 items (e.g., “active”, “determined”, “proud”; α=0.96, range=[.94, 0.96]), while negative affect was assessed using 10 items (e.g., “afraid”, “nervous”, “distressed”; α=0.94, range=[.92, 0.95]).

5.1.3. Analytic plan

5.1.3.1. Latent structure. We evaluated the same single-factor structure of the FoMO-3 used for trait-like FoMO with the same three items. We analysed the structure in terms of fit to the weekly data: RMSEA, an absolute fit index which is parsimonious in nature and sensitive to estimated parameters in the model [15]; SRMR, an absolute fit index which minimises the average Type I and Type II error rates [26]; and CFI, an incremental fit index which compares the fit of a hypothesised model with that of a baseline model (Bhale & Bedi, 2023; [12,24,57,

Table 5
Participant-level descriptive statistics.

Characteristic	*M* (*SD*) or %		Observed Range	Theoretical Range
Weekly FoMO (*N* = 250)				
Demographics				
Sex (% female)	71.60 %			
Race (% Chinese)	80.00 %			
Age	21.83	(1.76)	18–29	
Socioeconomics				
Monthly Household Income	3.20	(1.49)	1–6	1–6
Ladder Rank	6.02	(1.30)	2–9	1–10
Participation				
No. of Study Weeks Completed	12.07	(1.94)	2–13	2–13
Daily FoMO (*N* = 512)				
Demographics				
Sex (% female)	75.20 %			
Race (% Chinese)	80.10 %			
Age	22.24	(1.68)	19–30	
Socioeconomics				
Monthly Household Income	3.02	1.44	1–6	1–6
Ladder Rank	6.16	1.32	2–10	1–10
Participation				
No. of Study Days Completed	6.83	0.50	2–7	2–7

*Note. N*s refer to the number of participants.

114]). The models were considered good if they met these criteria: RMSEA≤.06 [57], SRMR≤.08, ([100] Shi & Maydeu-Olivares, 2020), and CFI>0.95 [11]. As participants were not required to complete all weeks of data collection, some participants had incomplete data. Such missing data was handled using full information maximum likelihood (FIML).

Given that strict measurement invariance across sex was established in Study 1, we can reasonably expect that the same measure, using the exact same items, would demonstrate invariance across sex as well. We were not able to formally test for measurement invariance across sex in this dataset due to an insufficient number of male participants to support meaningful analysis.

5.1.4. Measurement invariance over time

We tested for measurement invariance to ensure that the FoMO-3 operates in the same manner across time, specifically over weekly intervals across 13 weeks in the current study. Measurement invariance over time is held when participants interpret the items in relation to the underlying latent factor in the same way across all weeks [107]. That is, any change in item scores over time is fully attributable to a true change in the level of latent FoMO, rather than a change in how the scale is interpreted across time points.

Model comparisons were generally conducted in a similar manner to measurement invariance by sex, following the same four consecutive steps [71]. However, in order to account for the structure of the weekly diary data, we imposed two constraints: first, latent FoMO was allowed to correlate only across consecutive weeks (e.g., FoMO at week 2 correlated with FoMO at week 1 and week 3, but not with FoMO at week 4); and second, the residual error variances of each item of each week correlated with the residual error variances of the same item in the consecutive weeks only (e.g., error variances of item 1 at week 2 correlated with item 1 at week 1 and week 3, but not with item 1 at week 4). We modelled only the covariances between consecutive weeks to preserve the temporal order of the relationships and to reduce model complexity. Allowing values at all weeks to freely intercorrelate not only introduces high levels of computational load when estimating parameters, but more importantly, lacks theoretical grounding, as the temporal structure of the data follows a weekly structure. By focusing on consecutive weeks, we ensured that the relationships reflected meaningful patterns over time, without introducing unnecessary complexity to the structure of the model.

Following the imposition of this structure, we proceeded to test for measurement invariance using the same steps as Study 1. The first step of configural invariance was tested by only constraining the general factor structures to be the same (i.e., the same items loaded onto the same factor) across weeks. Next, weak invariance was tested by constraining factor loadings to equality across weeks, and if established, it showed that items did not vary in how representative they were of the latent construct over time. Strong invariance was tested by constraining factor loadings and item intercepts to equality across weeks. If strong invariance was held, we can conclude that any observed change in latent means over time reflects a true change in latent mean rather than how the participants interpret the response scale differently over time. Lastly, strict invariance was tested by constraining the factor loadings, item

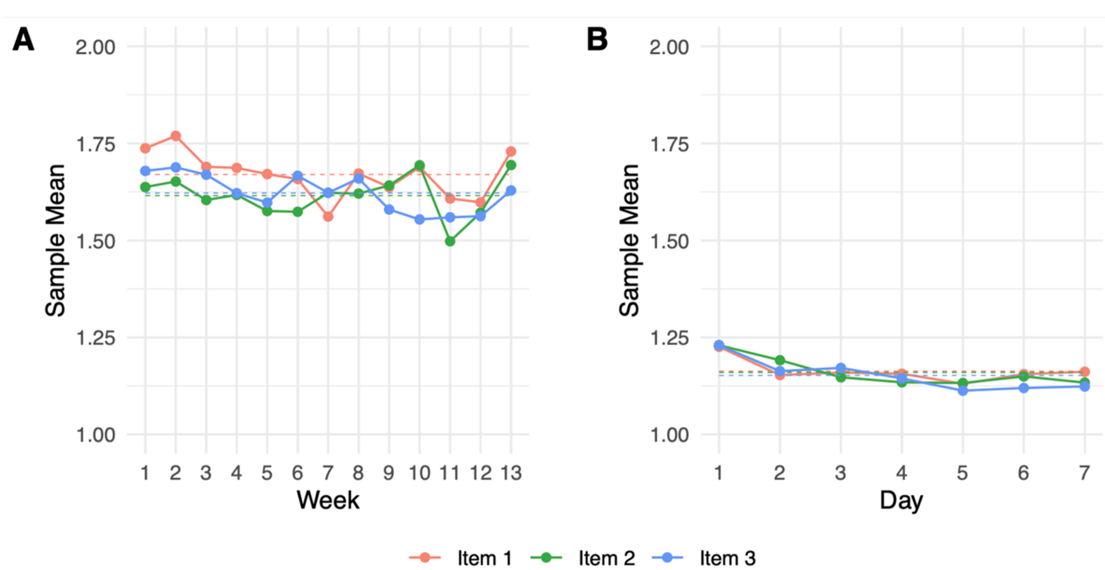

Fig. 5. Sample mean of each FoMO-3 item over time.
Note. The full theoretical range of the score on each item is 1–5. Dashed horizontal lines indicate the grand mean of all sample means over time. See Siepe et al. [101] for an in-depth explanation of this style of plot.

intercepts, residual variances, and residual covariances to be equal across weeks. If strict invariance is held, any observed changes in the item scores over time reflect true changes rather than being influenced by measurement error or other external factors that significantly change over time [107]. If strict invariance was achieved, we can confidently conclude that FoMO was measured consistently across time, that is, items reflect the FoMO construct consistently throughout the 13 weeks.

5.1.4.1. Homogeneity of latent means, variances, and covariances by time. After establishing measurement invariance, tests for homogeneity of latent means, variances, covariances, and means were conducted on the final model which passed measurement invariance testing. We explored whether the distributions (i.e., means and variances) were consistent at two separate time points (i.e., means and variances at week 1, and means and variances at week 2). For example, the variance in latent FoMO at week 1 and at week 2 were fixed to equality. Additionally, homogeneity tests of covariances revealed whether the covariances of latent FoMO between consecutive weeks were stable over time, in that the relationship of latent FoMO between week 1 and week 2 was the same as the relationship of latent FoMO between another set of consecutive weeks (e.g., week 12 and 13). Considering these two pieces of information concurrently allowed us to assess whether weekly FoMO exhibited temporal stability through the stability of absolute distribution and whether relationships of weekly FoMO between consecutive time points were stable.

We used the same criteria as measurement invariance testing (i.e., lower AIC, lower BIC, or ΔCFI<0.010). If the model with constrained latent means, variances, and covariances showed a better fit than the unconstrained models, we concluded that the distribution and temporal interrelationships of FoMO remained consistent over the weeks.

5.1.4.2. Psychometric properties. To ensure that the FoMO-3 is a reliable and valid measurement of weekly FoMO, we assessed its psychometric properties, namely internal consistency, average variance extracted, and temporal stability. The former two properties were measured in the same manner as trait FoMO and will therefore not be discussed in detail.

Temporal stability—often termed as test-retest reliability—was assessed by examining the correlation coefficients between the consecutive weeks from the final model. The strength of the relationships across time can be assessed using Pearson's r, where $r \geq .90$ suggests excellent temporal stability, $0.75 \leq r < 0.90$ suggests good temporal stability, $0.50 \leq r < 0.75$ suggests moderate temporal stability, and $r < 0.50$ suggests poor temporal stability [62]. As the current study aimed to examine a theoretically state-like construct that was meant to vary week-to-week, we expected low to moderate levels of temporal stability, and not high levels of temporal stability which would be more indicative of a trait-like construct.

5.1.4.3. Correlations with FoMO. We aimed to explore the relationship between FoMO and variables that vary over time, specifically positive and negative affect. Additionally, we investigated whether the correlations between FoMO and each type of affect differed across time by comparing two models: a constrained and an unconstrained model. The unconstrained model allowed all variables to correlate freely across time (e.g., the correlation between latent FoMO at week 1 and week 2 was not forced to be equal to the correlation between latent FoMO at week 6 and week 7). However, the intercorrelations between positive and negative affect within the same week (e.g., the correlation between positive affect at week 1 and negative affect at week 1) were constrained to equality across all weeks in both models, as the relationship between these constructs was assumed to remain constant. On the other hand, the constrained model forced all correlations across time to equality in order to test the assumption that all correlations among variables were consistent over time. For example, covariance between FoMO and positive affect at week 1 and the covariance between FoMO and positive affect at week 2 was forced to equality.

Subsequently, the two models were compared against each other using the same criteria as measurement invariance testing (i.e., lower AIC, lower BIC, or ΔCFI<0.010) to identify the model with a better fit. By comparing the fits of these two models, we were able to deduce relationships between the following pairs of variables: FoMO and positive affect, and FoMO and negative affect, in the same week. If the constrained model showed a better fit, the relationships between variables were generally consistent over time. For example, this means that the correlation between FoMO and positive affect in Week 1 is expected to be the same as the correlation between FoMO and positive affect in Week 7. Conversely, if the unconstrained model showed a better fit, the relationships between variables generally vary over time. For example, the correlation between FoMO and positive affect in Week 1 is expected to be different from the correlation between FoMO and positive affect in Week 7. Subsequently, using the model with the better fit, we interpreted the intercorrelations among FoMO, positive affect, and negative affect.

5.2. Results

5.2.1. Measurement invariance by time

Table 6 contains full information about the model fit and model comparison indices for the weekly version of the FoMO-3 measure. First, the configural invariance model displayed mediocre fit. Weak invariance was tested by comparing the configural invariance model against a new model with factor loadings constrained to equality across time; weak invariance was held based on all three model comparison indices. Next, strong invariance was tested by comparing the weak invariance model to a model with the additional equality constraint of item intercepts across the weeks, and strong invariance was held based on AIC and BIC. Afterward, we tested for strict invariance by comparing the strong invariance model to a model with the additional equality constraint of residual variances and covariances across the weeks. Strict invariance was not supported based on ΔCFI, AIC, or BIC values. Hence, the weekly FoMO-3 measure demonstrated strong invariance across time, indicating that, controlling for the latent levels of FoMO, the item intercepts of the measure are comparable over time, while the residual variances and covariances are not stable over time.

Table 6
Model fit indices and model comparison indices of weekly and daily FoMO models.

Model	RMSEA	SRMR	CFI	ΔCFI	AIC	BIC
Weekly FoMO (N = 250)						
Measurement Invariance						
Configural	.115	.473	.809	–	16,444	17,025
Weak (vs. Configural)	.113	.472	.808	.001	16,433	16,929
Strong (vs. Weak)	.111	.473	.806	.002	16,422	16,791
Strict (vs. Strong)	.111	.474	.789	.017	16,544	16,670
Homogeneity						
Variances and Covariances and Means (vs. Strong)	.111	.472	.801	.005	16,447	16,740
Daily FoMO (N = 512)						
Measurement Invariance						
Configural	.109	.286	.876	–	8093	8461
Weak (vs. Configural)	.107	.287	.872	.004	8115	8433
Strong (vs. Weak)	.103	.287	.870	.002	8116	8358
Strict (vs. Strong)	.123	.288	.785	.085	8776	8878
Homogeneity						
Variances and Covariances and Means (vs. Strong)	.103	.287	.863	.007	8160	8359

Note. Ns refer to the number of participants.

5.2.2. Homogeneity of latent variances, covariances, and means by time

Using the strong invariance model, we then conducted homogeneity tests of latent variances, covariances, and means by time by comparing the strong invariance model to an alternative model with an added constraint of latent variances, covariances, and means constrained to equality across weeks. At this step, we found evidence for homogeneity in latent variances, covariances, and means based on ΔCFI and AIC (Table 6). Hence, we concluded that the distributions and relationships of the FoMO latent construct were stable over time. The final factor structure is shown in Fig. 6.

5.2.3. Psychometric properties

We computed tau-equivalent and congeneric internal consistencies with median values of 0.933 (range=[.910, 0.956]) and 0.818 (range=[.726, 0.987]) respectively. At all time points, both estimates exceeded 0.70, which denotes good internal consistency at each time point (Table 7). The median value as well as lowest value of the average variance extracted (AVE=0.725; range=[.649, 0.859]) exceeded the acceptable threshold of 0.50, indicating that FoMO accounted for more than half of the variance associated in item scores over the weeks.

Since the homogeneity tests passed when latent means, variances and covariances were constrained to equality across weeks, the distributions of FoMO and the correlations between consecutive weeks were stable over time.[2] An examination of the week-to-week correlations of latent FoMO (which were constrained across time) revealed that the relationship of FoMO between consecutive weeks was r=0.423, which suggests poor temporal stability (Table 7).

5.2.4. Correlations with FoMO

First, we found that the constrained model (AIC=30,178, BIC=30,749, CFI=0.681) displayed a better fit compared to the unconstrained model (AIC=30,183, BIC=30,838, CFI=0.682). This suggests that there was no evidence for differences in the relationships between FoMO and affect over time. From the constrained model, significant zero-order correlations were found between FoMO and positive in the same week, and between FoMO and negative affect in the same week. Specifically, FoMO was negatively related to positive affect to a trivial extent with a median value of r=–.022 (range=[–.024, –.020]; Fig. 7), such that in weeks where FoMO was higher, positive affect was slightly lower. FoMO was positively related to negative affect of the same week to a very small extent with a median value of r=0.067 (range=[0.062, 0.072]; Fig. 7), such that high levels of FoMO were associated with high levels of negative affect during the same week.

6. Study 3: daily FoMO

6.1. Method

6.1.1. Sample and design

The sample for the study of daily FoMO was drawn from two independent but methodologically similar daily diary projects comprising an initial total of 514 participants (see [46,53,70,80]). Participants were allowed to respond to the survey each day from 8 pm. till 3 a.m. the following day and were provided compensation of up to SGD$70 for their participation. Following the baseline questionnaire, participants were instructed to complete seven consecutive daily surveys. Participants were included if they completed at least two days of daily data collection, reducing the sample from 514 to 512 participants. This resulted in a total of 3498 observations of daily data available for analysis. A summary of descriptive statistics is reflected in Table 5 and Fig. 5b

6.1.2. Daily measures

6.1.2.1. Daily FoMO. Daily FoMO was measured using the same three-item version of the Fear of Missing Out Scale [85]. Similar to the weekly diary study, the original scale was adapted to include a 24-hour time-frame (i.e., "In the past 24 h"), at the beginning of each item to guide the specificity of responses of the daily diary study. Participants rated their agreement to these items on a five-point scale (1=*Not at all true of me*, 5=*Extremely true of me*) daily.

6.1.2.2. Daily positive and negative affectivity. Daily positive and negative affectivity were measured using the Daily Distress Scale [79]. Participants were asked how frequently in a day they experienced the following positive and negative emotions on a 5-point Likert scale (e.g., 0=*None of the time*, 4=*All of the time*). Positive affect was assessed using 13 items (e.g., "cheerful", "extremely happy", "full of life"; α=0.97, range=[.97, 0.98]), while negative affect was assessed using 14 items (e.g., "nervous", "hopeless", "lonely"; α=0.96, range=[.95, 0.96]).

6.1.3. Analytic plan

The analytic plan of Study 3 was identical to that of Study 2, barring that instead of an investigation of weekly FoMO across 13 weeks, we instead investigated daily FoMO across 7 days.

6.2. Results

6.2.1. Measurement invariance by time

Table 6 contains full information about the model fit and model comparison indices for the daily version of the FoMO-3 measure. Similar to the findings from the weekly version, the configural invariance model displayed mediocre fit, weak invariance was held, and strong invariance was held. In line with the findings from the weekly version, strict invariance was not held. Hence, the daily FoMO-3 measure demonstrated strong invariance across time, indicating that, controlling for the latent levels of FoMO, the item intercepts of the measure are comparable over time, while the residual variances and covariances are not stable over time.

6.2.2. Homogeneity of latent variances, covariances, and means by time

Using the strong invariance model, we then conducted homogeneity tests of latent variances, covariances, and means by time by comparing the strong invariance model to an alternative model with an added constraint of latent variances, covariances, and means constrained to equality across days. At this step, we found insufficient evidence for homogeneity in latent variances, covariances, and means (Table 6). Hence, we concluded that the distributions and relationships of the FoMO latent construct were not stable over time—a contrast to the results found for weekly FoMO, which we return to in the Discussion. The final factor structure for daily FoMO is shown in Fig. 8.

6.2.3. Psychometric properties

We computed tau-equivalent and congeneric internal consistencies with median values of 0.889 (range=[.834, 0.928]) and 0.832 (range=[.752, 0.925]) respectively. At all time points, both estimates exceeded 0.70, which denotes good internal consistency (Table 7). The median value as well as lowest value of the average variance extracted (AVE=0.692; range=[.616, 0.811]) exceeded the acceptable threshold of 0.50, indicating that FoMO accounted for more than half of the variance associated in item scores over the days. These results are in line with those from weekly FoMO.

Since the homogeneity tests did not pass when latent means, variances and covariances were constrained to equality across weeks, the

[2] Note that this stability is a property of the sample and not of individual participants. That is, the result does not imply that the level of weekly FoMo for an individual participant is stable across time. Rather, it implies that, as a whole sample, the sample means, sample variances, and sample covariances of weekly FoMO are stable over time.

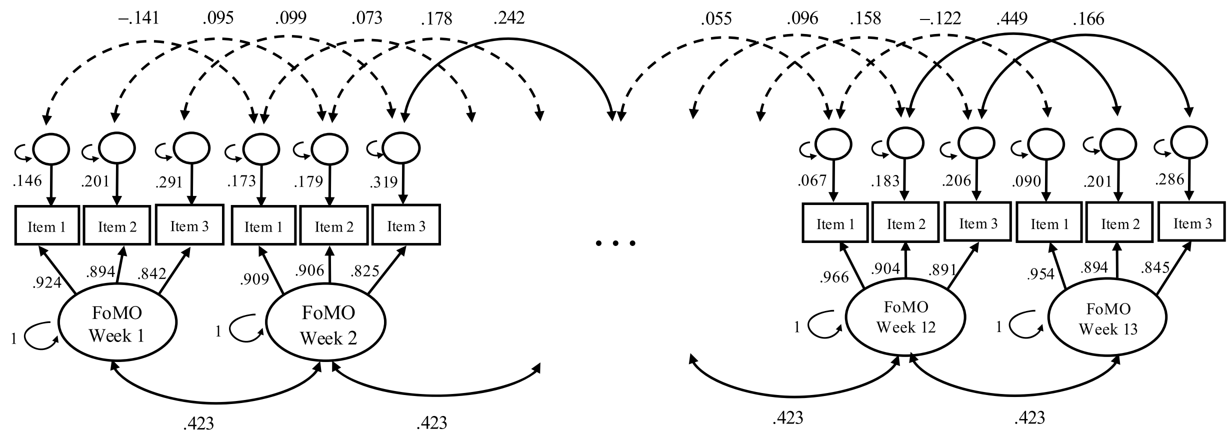

Fig. 6. Factor structure of weekly FoMO.

Note. $N_{participants}$=250. All values are standardised, including variances which are standardised to 1. Factor loadings, as well as latent variances, covariances, and means, were constrained to equality over time. However, they may appear unconstrained due to the failure of strict invariance constraints. Residual variances and covariances, on the other hand, were not constrained across time.

Table 7
Estimates of psychometric properties of the weekly and daily FoMO measures.

Time	Temporal Stability (*r*) with respect to Previous Time Point	Tau-Equivalent Internal Consistency (α)	Congeneric Internal Consistency (ω)	Average Variance Extracted (AVE)
Weekly FoMO (*N* = 250)				
Week 1	–	.930	.787	.691
Week 2	.423	.919	.818	.706
Week 3	.423	.931	.824	.725
Week 4	.423	.933	.854	.754
Week 5	.423	.924	.987	.859
Week 6	.423	.945	.767	.692
Week 7	.423	.910	.930	.790
Week 8	.423	.935	.747	.663
Week 9	.423	.956	.726	.668
Week 10	.423	.951	.808	.737
Week 11	.423	.917	.925	.795
Week 12	.423	.943	.893	.804
Week 13	.423	.937	.729	.649
Daily FoMO (*N* = 512)				
Day 1	–	.834	.842	.634
Day 2	.317	.889	.752	.616
Day 3	.476	.904	.823	.692
Day 4	.228	.886	.821	.670
Day 5	.494	.885	.851	.693
Day 6	.184	.911	.832	.708
Day 7	.549	.928	.925	.811

*Note. N*s refer to the number of participants.

distributions of FoMO and the correlations between consecutive weeks were not stable over time. An examination of the day-to-day correlations of latent FoMO across time revealed that the relationship of FoMO between consecutive days ranged from poor (r=0.184) to moderate (r=*0*.549), with a median value of r=0.397, which suggests overall poor temporal stability (Table 7).

6.2.4. Correlations with FoMO

In line with the findings for weekly FoMO, when investigating the relationships between daily FoMO and daily affect, we found that the constrained model (AIC=23,899, BIC=24,323, CFI=0.692; Fig. 9) displayed a better fit compared to the unconstrained model (AIC=23,896, BIC=24,371, CFI=0.693), suggesting that there was no evidence for differences in the relationships between FoMO and affect over time. Drawing from the constrained model, we found that on days where FoMO was higher, positive affect was significantly lower to a trivial extent (median r=–.045, range=[–.051, –.039]; Fig. 9) and negative affect was significantly higher to a very small extent (median r=0.061, range=[.047, 0.070]; Fig. 9).

7. Discussion

The proliferation of social media has significantly amplified individuals' awareness of missed experiences [52,59], thereby intensifying the phenomenon of Fear of Missing Out (FoMO). FoMO has emerged as a pervasive negative state with well-documented implications for decision-making processes [76,94,110] and overall mental well-being [10,13,34,103]. Given the widespread prevalence and potential consequences of FoMO, developing psychometrically robust and efficient tools for its assessment is essential, particularly for time-intensive studies requiring repeated measurements. Hence, the present research aimed to develop and validate a shortened version of the FoMO-10 scale, designed to capture the construct efficiently across various temporal frames. Specifically, we examined the performance of the newly-developed FoMO-3 in terms of measurement invariance across sex and time, internal consistency, and temporal stability.

7.1. Summary of findings

The Pilot Study enabled the development of a shortened three-item measure of FoMO that preliminarily demonstrated promising psychometric properties, which were further evaluated in the subsequent studies. We found that the correlation between the mean score of the FoMO-10 and the mean score of the FoMO-3 was positive and nearly perfect (r=0.912, 95 % CI=[.877, 0.937]), indicating that the shortened scale is very closely related to the original version in capturing the underlying construct. This result is particularly desirable, given that our aim of developing a shortened scale still preserves the conceptual and psychometric integrity of the original measure while providing a more brief and efficient instrument for use, especially in time-intensive studies.

7.2. Measurement invariance and temporal stability

Study 1 provided strong evidence supporting the use of the FoMO-3 as a trait-like measure. The FoMO-3 displayed high inter-item consistency and an acceptable AVE score, exhibiting good psychometric

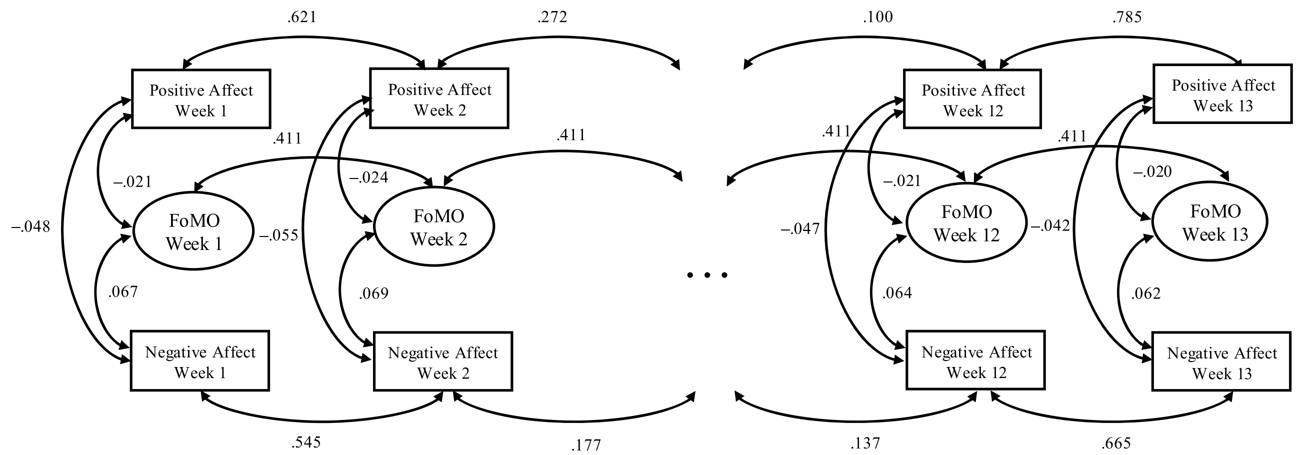

Fig. 7. Zero-order correlations of weekly FoMO with affect.
Note. $N_{\text{participants}}$=250. All values are standardised, including variances which are standardised to 1. Correlations between positive and negative affect were constrained over time. The three manifest variables of FoMO are not shown.

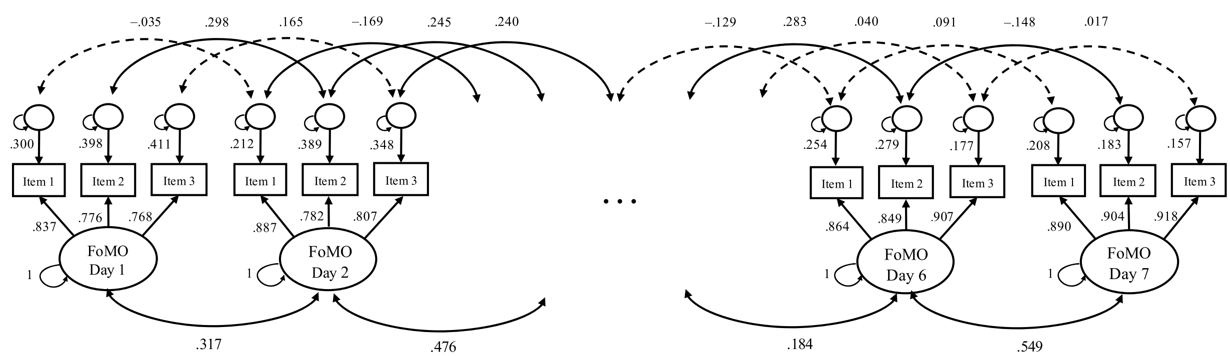

Fig. 8. Factor structure of daily FoMO.
Note. $N_{\text{participants}}$=512. All values are standardised, including variances which were standardised to 1. Factor loadings, as well as latent variances, covariances, and means, were constrained to equality over time. However, they may appear unconstrained due to the failure of other constraints. Residual variances and covariances, on the other hand, were not constrained across time.

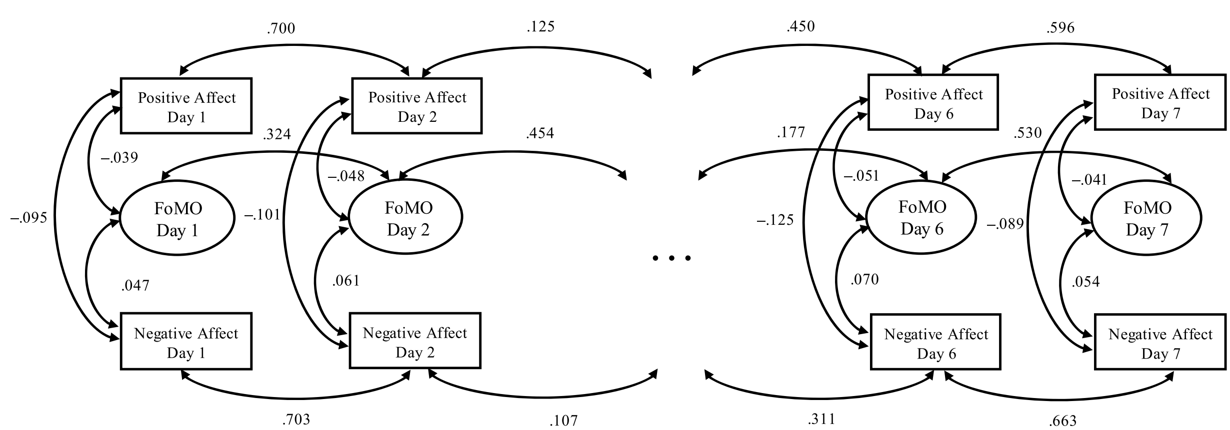

Fig. 9. Zero-order correlations of daily FoMO with affect.
Note. $N_{\text{participants}}$=512. All values are standardised, including variances which are standardised to 1. Correlations between positive and negative affect were constrained over time. The three manifest variables of FoMO are not shown.

properties at the trait level across both males and females. Additionally, FoMO-3 achieved strict invariance across sex, implying that FoMO was measured identically across males and females using the three items. Given that the FoMO-3 and FoMO-10 are almost perfectly correlated, the difference between the invariance results found in Study 1 and in the existing literature of FoMO-10 likely reflects differences in how the construct is represented at the item level. The FoMO-10 contains more heterogeneous items that tap into context-specific expressions of FoMO (e.g., missing out on social events or online interactions), which increases the likelihood of subtle sex-related differences in interpretation or endorsement. By contrast, the FoMO-3 comprises highly central, broadly worded items that reduce opportunities for such bias and yield a more parsimonious representation of the construct. Relatedly, strict invariance becomes more difficult to achieve as the number of items increases, since each additional item carries the potential for non-invariance. Thus, while both scales assess the same underlying construct, the brevity and generality of the FoMO-3 may explain why it achieved strict invariance across sex, in contrast to prior findings with the FoMO-10. Taken together, the results of these tests fully support the FoMO-3 as a psychometrically sound measure, suitable for assessing trait FoMO across sex. Further, homogeneity in latent means and variances was found across sex as well, which indicated that in general, males and females have the same average level and spread of FoMO. Future research should seek to replicate these findings using a more balanced distribution of males and females within a single sample.

Building on the promising results of FoMO-3 as a psychometrically sound trait-like measure, Studies 2 and 3 explored the validity of the FoMO-3 for assessing more state-like aspects of FoMO, examining weekly and daily fluctuations respectively. For state assessments, demonstrating temporal measurement invariance was crucial, as it ensures that observed changes in FoMO reflect true variations in the underlying construct rather than artifacts resulting from varying interpretations of the scale over time. Study 2 confirmed strong invariance of weekly FoMO scores across 13 time points separated by 1-week intervals, indicating consistent interpretations of item meanings across weeks. Importantly, we also demonstrated homogeneity in latent means, variances, and covariances across weeks, indicating stability in the distribution and interrelations of weekly FoMO. Consistent with theoretical expectations of the state-like nature of weekly FoMO which inherently involves fluctuations rather than consistency over time, the temporal stability across weeks was notably low. In Study 3, daily measurements similarly achieved strong measurement invariance, suggesting consistent item interpretation across consecutive days. However, unlike weekly assessments, homogeneity in latent means, variances, and covariances was not maintained over consecutive days. The distributions and interrelations of daily FoMO thus varied significantly from day to day, further supporting the notion that FoMO assessed daily is even more volatile and sensitive to contextual factors. Again, poor temporal stability was anticipated, reflecting the highly transient and context-dependent nature of daily FoMO experiences. Taking these findings in sum, despite capturing state-like aspects, weekly FoMO may inherently share more stable, trait-like characteristics than daily FoMO, which is more sensitive to immediate situational contexts.

7.3. Reliability and correlates of FoMO

Importantly, across all assessments, the FoMO-3 consistently demonstrated high internal consistency, with both tau-equivalent and congeneric estimates remaining well above established thresholds. This consistent reliability across trait, weekly, and daily measurements provides robust evidence that the three items collectively and coherently capture the core construct of FoMO, irrespective of the temporal resolution. Thus, researchers can confidently employ the FoMO-3 in diverse study designs, knowing that the internal coherence of the measure is maintained across varying assessment intervals.

Explorations into correlations between FoMO and relevant variables yielded important insights into its nomological network. Consistent with prior research (e.g., [8,37,38,116]), trait FoMO was positively associated with neuroticism, negative affect, and smartphone addiction, suggesting that individuals prone to negative emotional states and problematic smartphone usage experience heightened FoMO. These findings reinforce the conceptualisation of trait FoMO as a relatively stable psychological disposition connected to enduring personality and behavioural tendencies. In contrast, state-like FoMO assessments (weekly and daily) focused primarily on transient emotional experiences, notably positive and negative affect, where higher FoMO was consistently linked with decreased positive affect and increased negative affect.

7.4. Implications and future directions

While the FoMO-3 demonstrated robust psychometric validity at each separate time point and measurement occasion, its validity when tracking the temporal dynamics of FoMO over extended periods is nuanced. Although strong measurement invariance was achieved, indicating that participants interpret items consistently across time, the temporal instability suggests that FoMO, particularly daily, is a highly dynamic emotional state sensitive to changing contextual and affective conditions. This finding underscores the necessity of carefully selecting measurement intervals tailored to research questions regarding FoMO's temporal characteristics. The findings from the current work suggest that expanding the use of FoMO-3 to time-intensive study designs, such as experience sampling methods, could capture the immediacy and context-specific nature of FoMO more accurately, overcoming potential limitations associated with retrospective assessments.

8. Conclusion

Together, these three studies collectively validate the structure and measurement invariance of the FoMO-3 across sex and various temporal contexts, supporting its internal consistency and validity. A key contribution of this work is the demonstration that a brief, three-item scale effectively captures the construct of FoMO without sacrificing psychometric rigour, striking an optimal balance between brevity (as seen in single-item measures; [91]) and comprehensiveness (e.g., [85]). Thus, the FoMO-3 emerges as particularly suitable for intensive longitudinal designs where brevity is crucial.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used OpenAI's ChatGPT in order to improve the readability, fluency, and clarity in some parts of this manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Eva M. Hisham: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Andree Hartanto:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Nadyanna M. Majeed:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The relevant materials, data, and code required to reproduce our analyses have been made publicly available on ResearchBox (#3841; https://researchbox.org/3841).

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